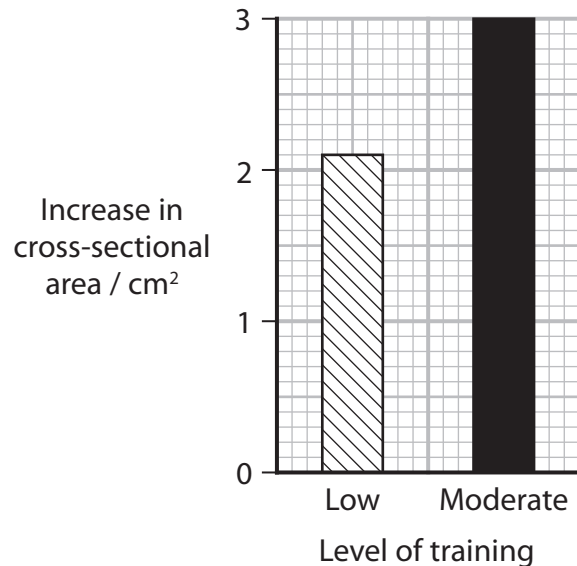
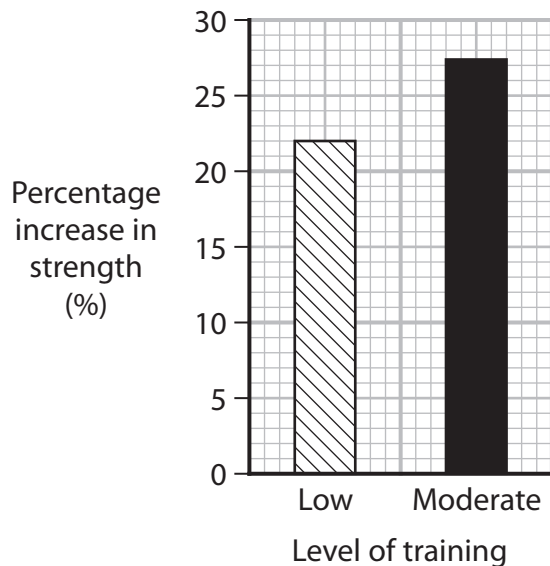


2 Many athletes train to improve muscle strength.

In an experiment, the effect on a muscle of 12 weeks of low or moderate training was investigated.

The graphs show the effect of 12 weeks of training on muscle strength and muscle cross-sectional area.



(a) The muscle strength of the athlete before moderate training was 12.5 a.u.

Calculate the muscle strength of the athlete after 12 weeks of moderate training.

Give your answer to 3 significant figures.

(3)

Answer

- (b) The types of muscle fibres present in samples taken before the training and after 12 weeks of low and moderate training were determined.

Fibre type	Percentage of fibres (%)		
	Before training	After 12 weeks of training	
		Low	Moderate
Slow twitch	44	44	44
Fast twitch	6	2	1
Intermediate	50	54	55

Intermediate fibres have structural properties between those of slow and fast twitch fibres.

- (i) Describe how the structure of an intermediate fibre will differ from the structure of a slow twitch fibre.

(3)

[illegible]

(ii) Comment on the effect of exercise on this muscle.

Use the information in the graphs and the table to support your answer.

(3)

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(Total for Question 2 = 9 marks)